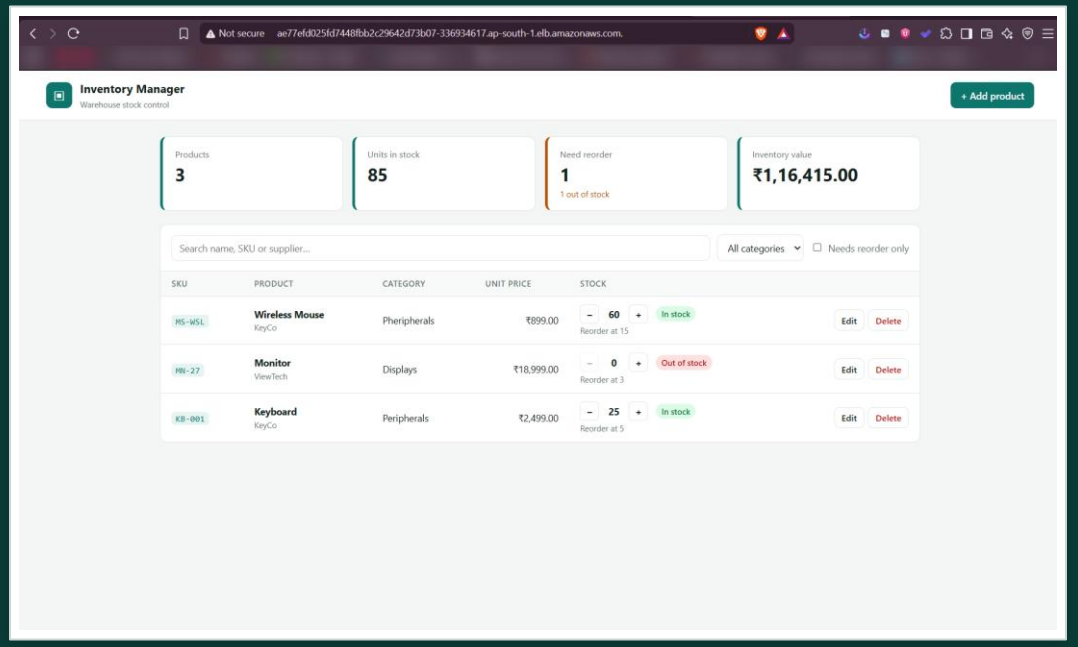
Inventory Manager

From git push to a monitored Kubernetes deployment on AWS



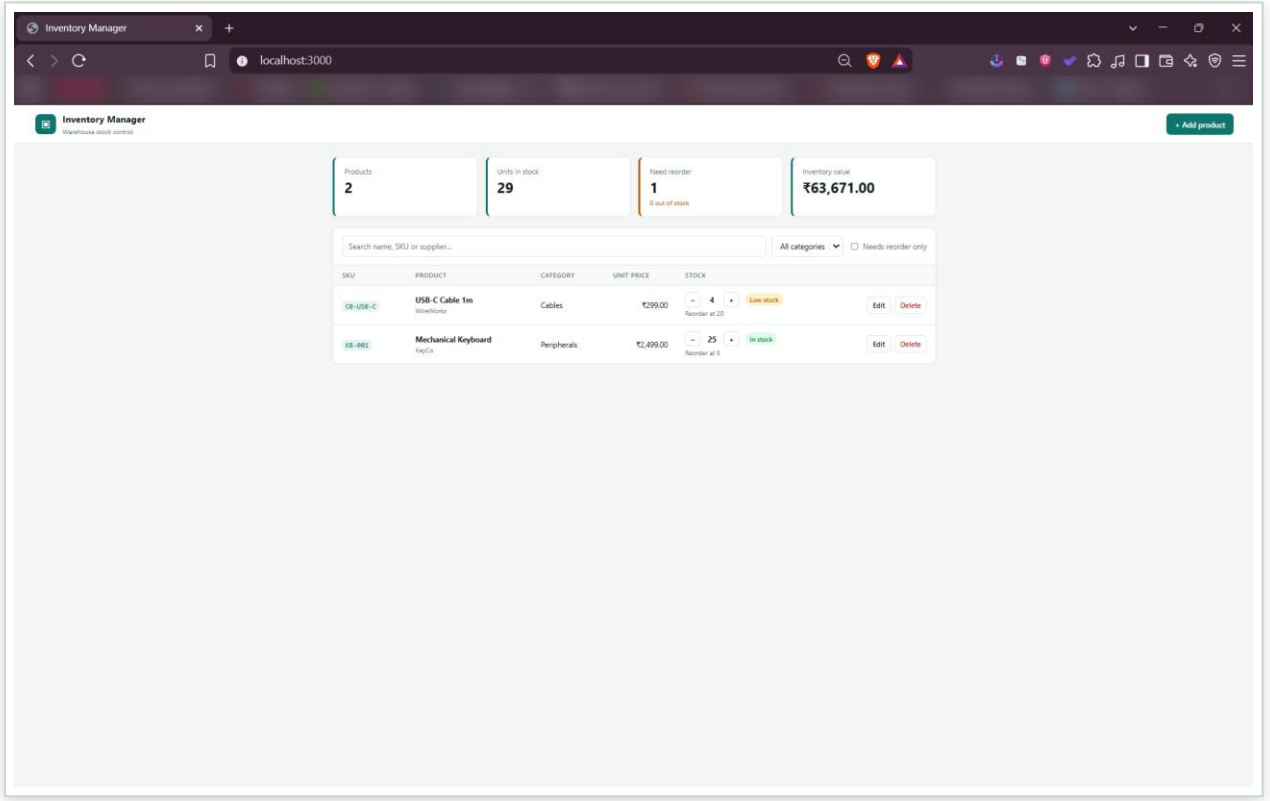
DevOps Capstone · Mohit Sagar · October 2026

What the app does

1. **Track products by SKU**
   Category, supplier, unit price and reorder level

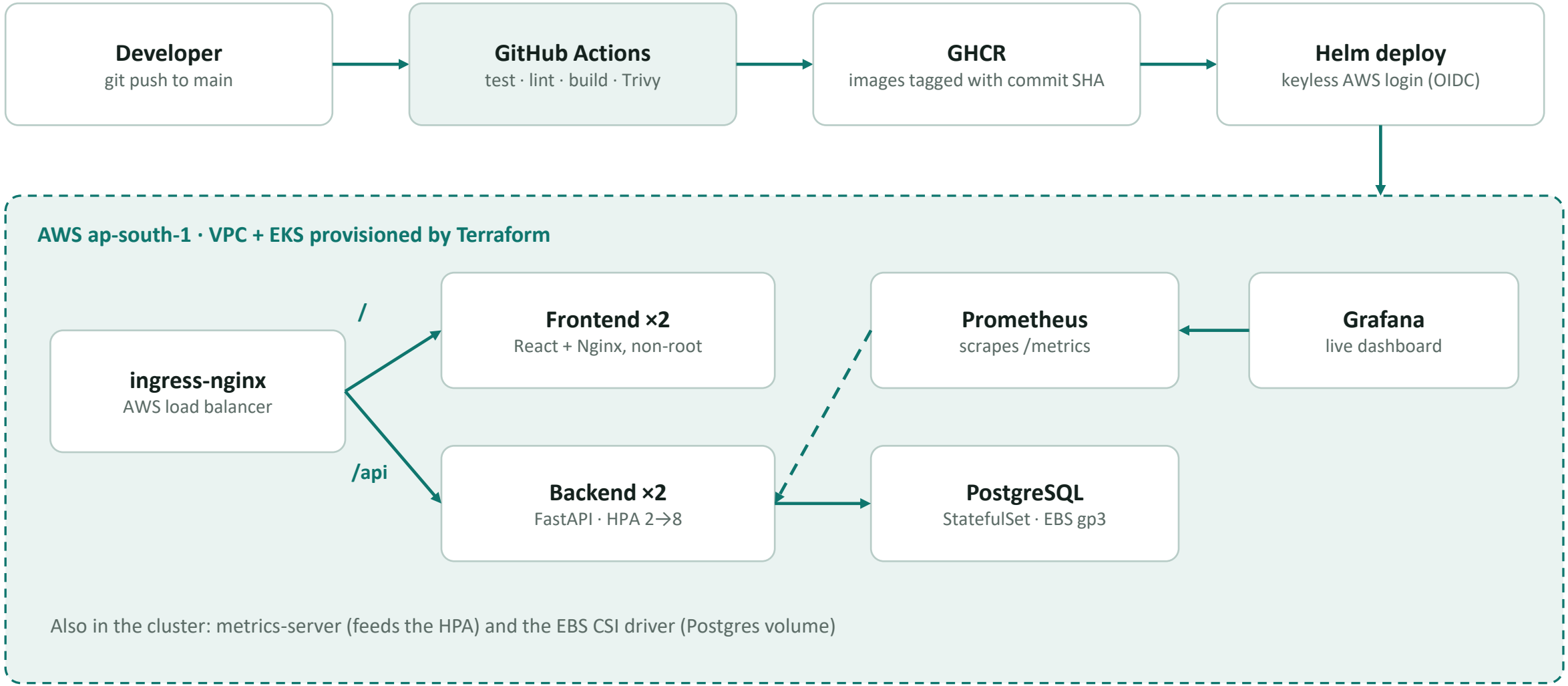
2. **Receive and ship stock**
   One-click + / −; stock can never go below zero

3. **Reorder alerts and value**
   Low / out-of-stock badges and total inventory value



React · FastAPI · PostgreSQL · Alembic · 10 REST endpoints

Architecture



CI/CD on every push to main

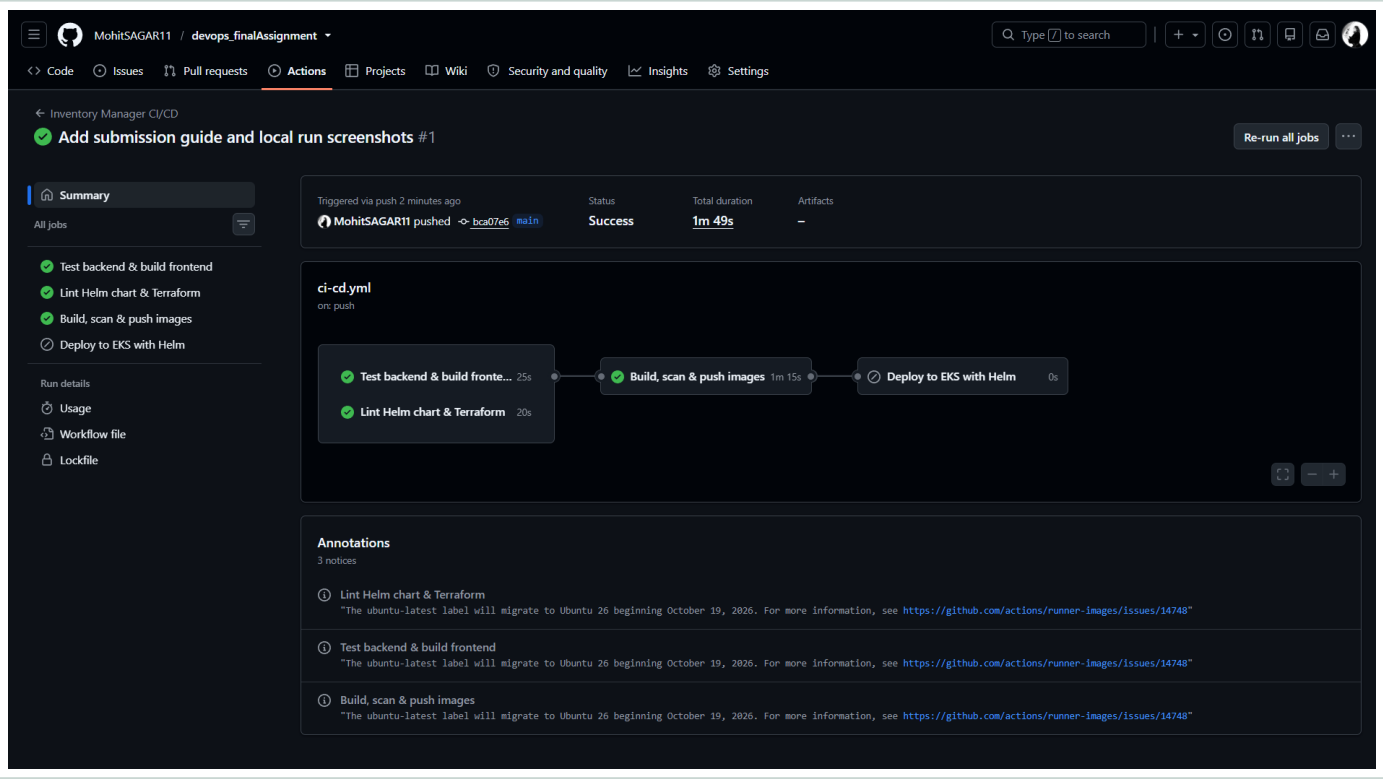
M5 · CI/CD

1 Test
pytest (19) + frontend build

2 Lint
helm lint + terraform validate

3 Build · Scan · Push
SHA tags, Trivy gate, GHCR

4 Deploy
OIDC → helm upgrade on EKS



Why the commit SHA?

Every running pod traces back to the exact commit that built it, and a rollback is just redeploying an older tag.

Fails fast

A failing test or a HIGH/CRITICAL CVE stops the pipeline before anything reaches the registry or the cluster.

Security gate: Trivy

M6 · DevSecOps

Before
starlette 0.41.3: CVE-2025-62727 (Range-header DoS)
nginx 1.27 base: fixable CVEs in libxml2, musl, zlib

After
Upgraded FastAPI / Starlette 1.8
nginx-unprivileged 1.30 + apk upgrade

0

HIGH / CRITICAL CVEs in both images

Supply chain
Every action pinned to a commit SHA: trivy-action tags were hijacked in March 2026 (CVE-2026-33634).



Infrastructure as Code on AWS

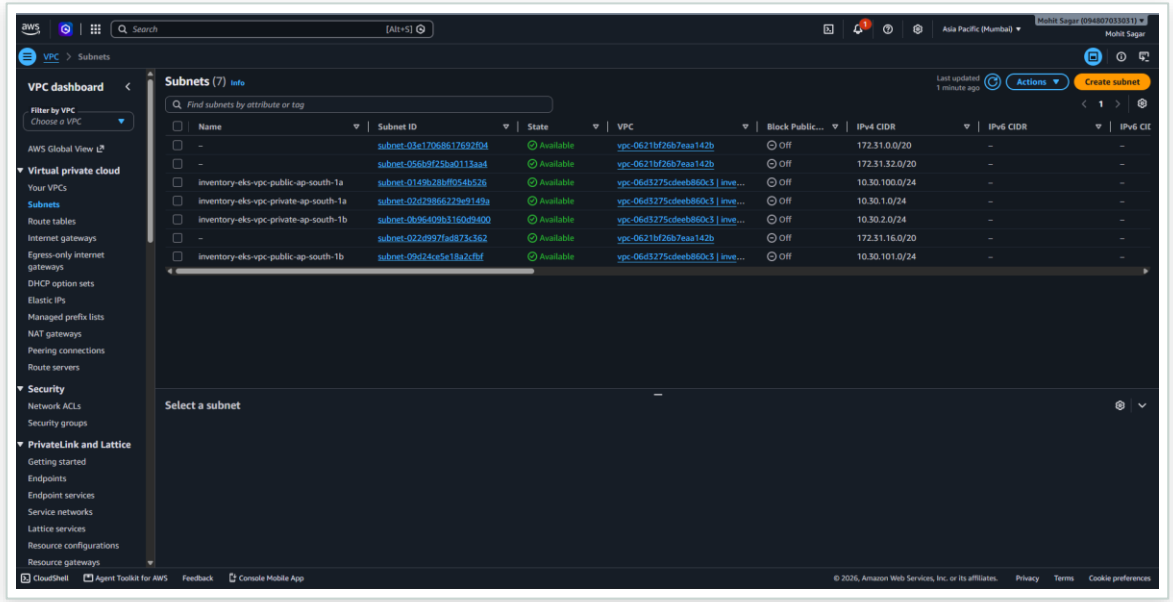
M7 · Terraform

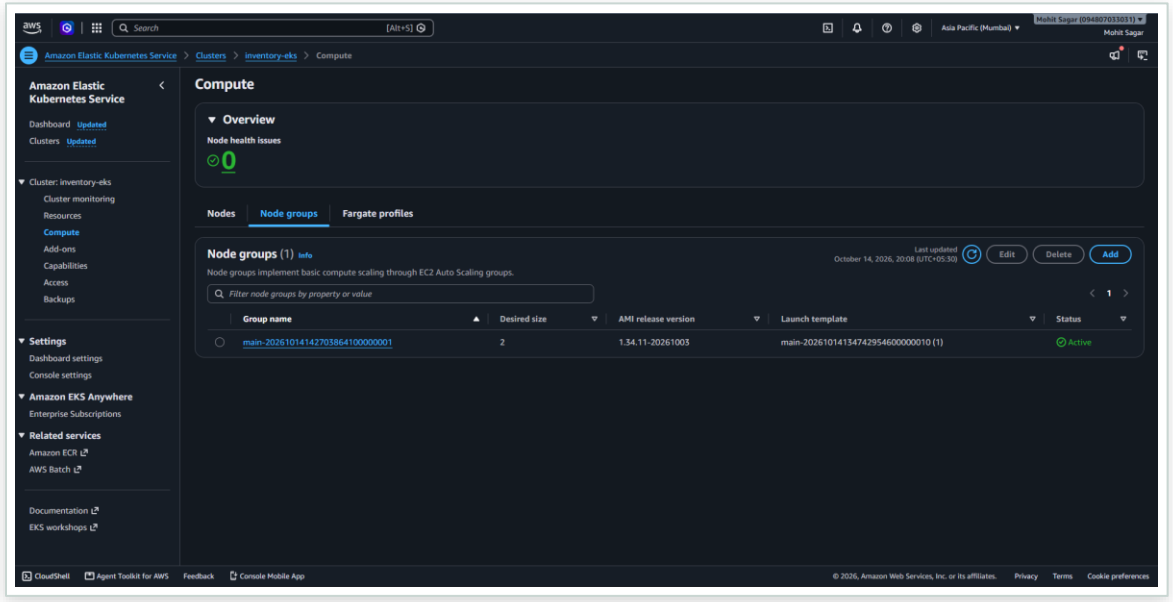
66
resources from terraform apply

4 subnets
2 public + 2 private across 2 AZs

2 nodes
m7i-flex.large, EKS 1.34

66
destroyed afterwards: no runaway cost





Kubernetes + Helm

Deployments
backend + frontend, 2 replicas

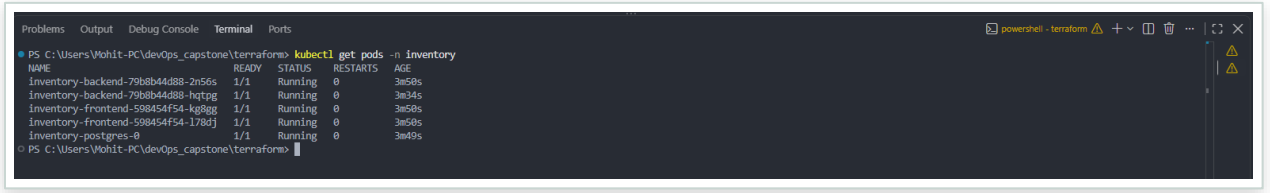
StatefulSet
Postgres on EBS gp3

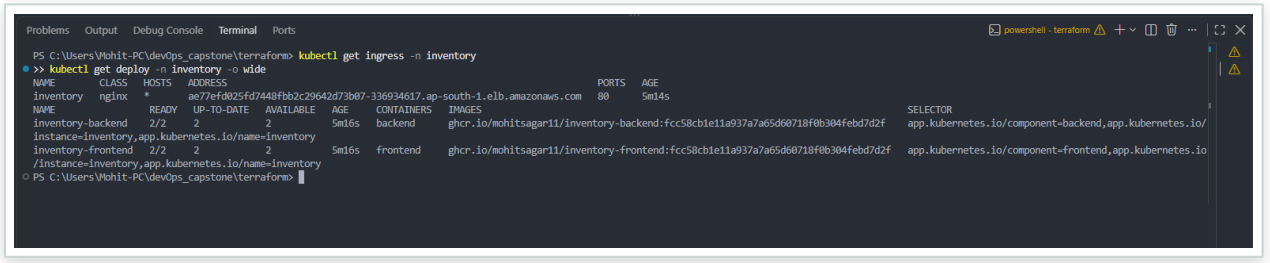
Services
ClusterIP for all three

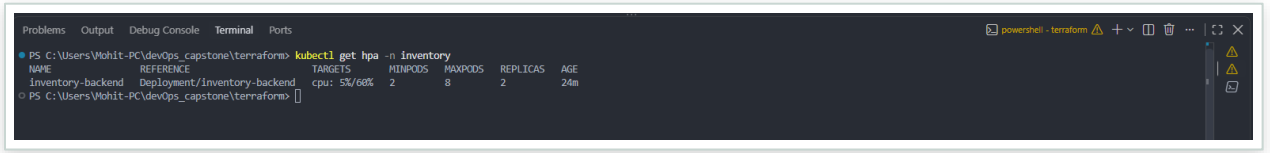
Ingress
/ → frontend, /api → backend

HPA
backend 2→8 at 60% CPU

ServiceMonitor
+ Grafana dashboard



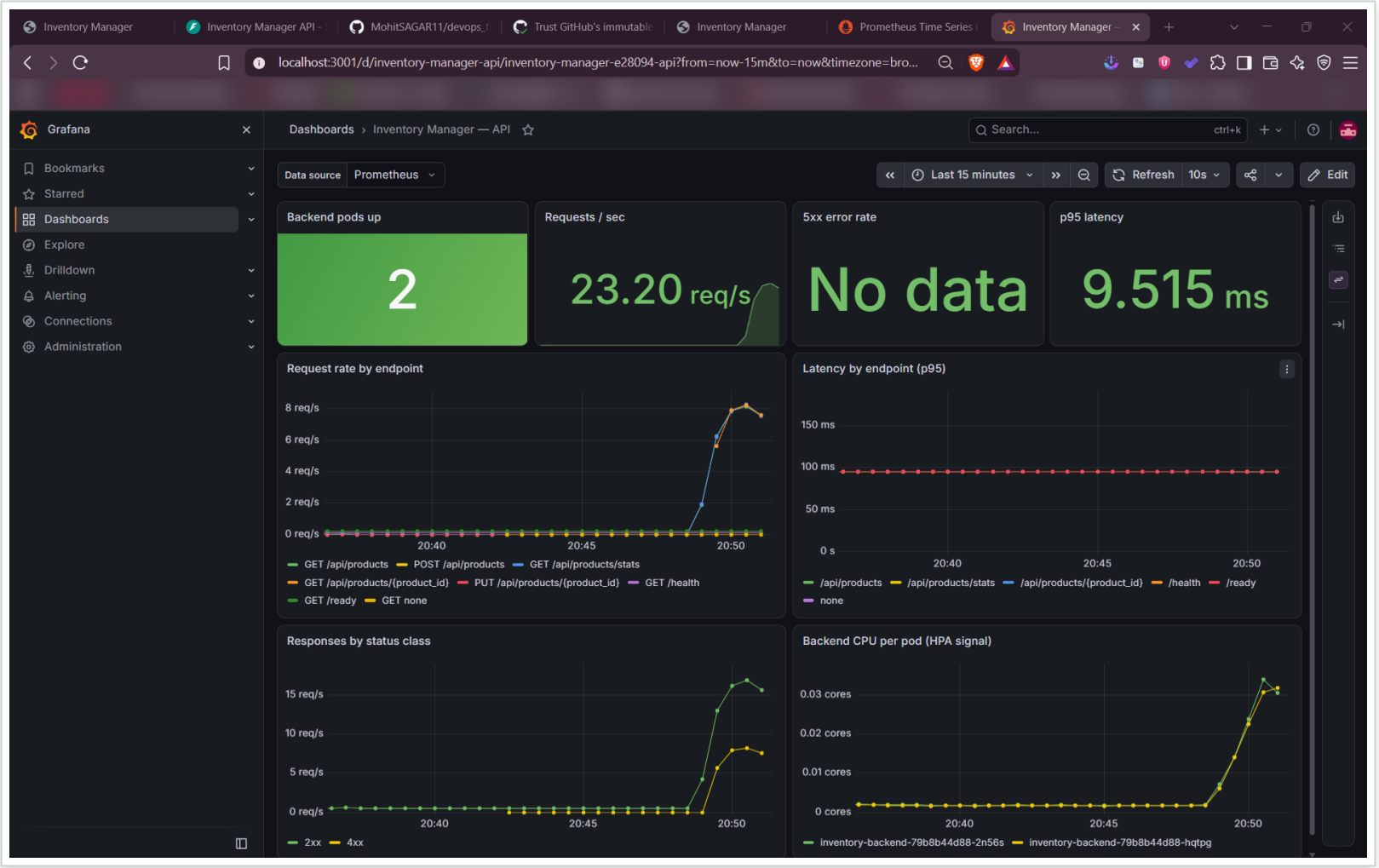




One chart, three values files (default, dev, prod).
CI deploys every push with helm upgrade --install.

Observability

M9 · Monitoring



2 / 2

backend targets UP in Prometheus

23 req/s

during the load test

9.5 ms

p95 latency under load

Quality gates before anything ships

M2 · M4

19

pytest cases, isolated SQLite DB

Non-root

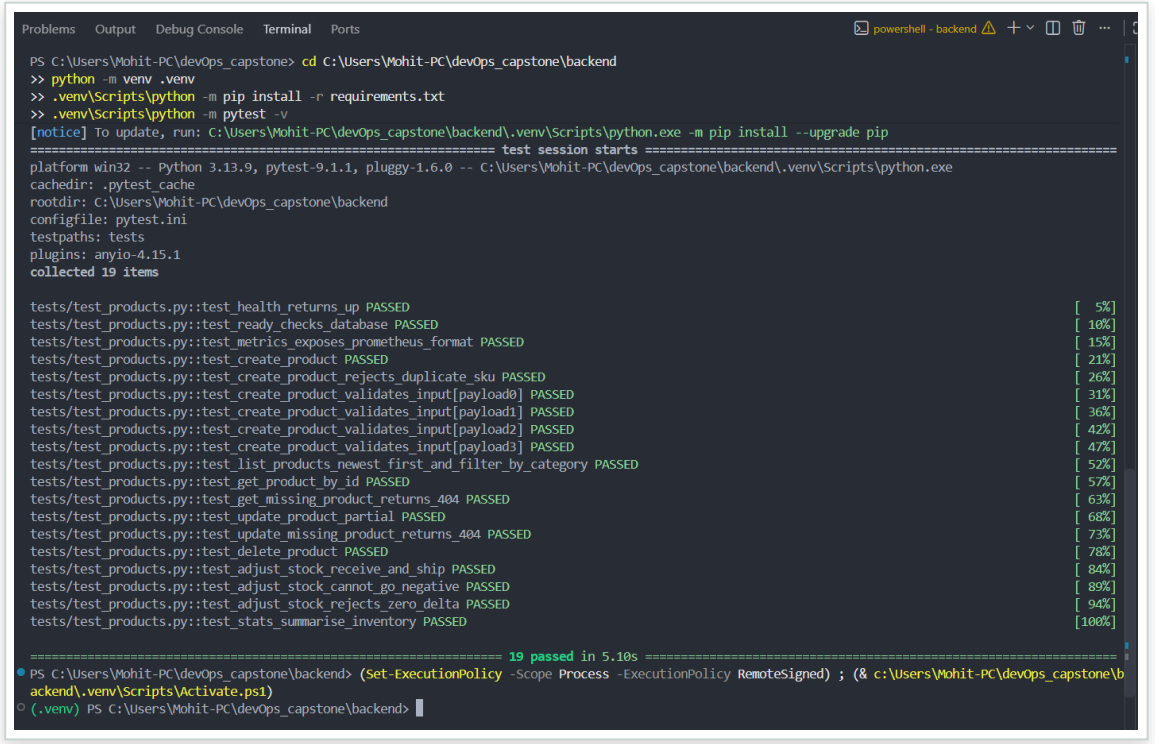
uid 10001 backend, uid 101 Nginx

Multi-stage frontend image

Node builds the bundle; only static files ship in unprivileged Nginx.

One-command local stack

docker compose up --build starts Postgres, backend and frontend with health checks.



Problems solved along the way

Trivy blocked the first release

Fix: Upgraded Starlette and the nginx base image

Two pods raced on DB migrations

Fix: Postgres advisory lock around alembic upgrade

Backend started before Postgres

Fix: TCP health check and a wait-for-postgres init container

AWS Free plan rejected t3.medium

Fix: Switched worker nodes to m7i-flex.large

CI deploy: AssumeRole denied

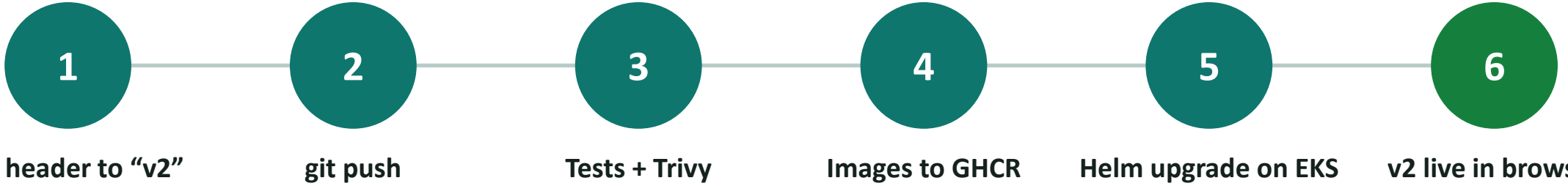
Fix: Trusted GitHub's immutable OIDC subject

Uppercase username invalid in GHCR

Fix: Lowercased the owner in image names

Live demo: commit to production

M10 · Demo



≈ 3 min

from git push to the new version serving traffic

Recorded walkthrough

demo/demo.mp4 · embedded in the README

github.com/MohitSAGAR11/devops_finalAssignment

Thank you

Questions?

github.com/MohitSAGAR11/devops_finalAssignment

27
commits

19
tests

0
HIGH/CRITICAL CVEs

66 / 66
AWS resources created / destroyed